EXPERIMENT-21 Towers of Hanoi

Aim

To write a Prolog program to solve the **Towers of Hanoi** problem.

Objectives

- To understand recursion in Prolog.
- To implement the Towers of Hanoi problem using recursive rules.
- To generate the sequence of moves required to transfer disks.

Algorithm

1. If there is only one disk, move it directly from source to destination.
2. For N > 1:
   - Move N-1 disks from source to auxiliary.
   - Move the largest disk from source to destination.
   - Move N-1 disks from auxiliary to destination.
3. Repeat recursively until all disks are moved.

Flowchart

```
    START
      |
      v
 Read number of disks N
      |
      v
   Is N = 1?
   /     \
  Yes     No
   |       |
```

```
    v         v
 Move disk   Move N-1 disks
    |        Source -> Auxiliary
    |          |
    |          v
    |       Move largest disk
    |          |
    |          v
    |       Move N-1 disks
    |       Auxiliary -> Destination
    |          |
    +----------+
         |
         v
       STOP
```

Program

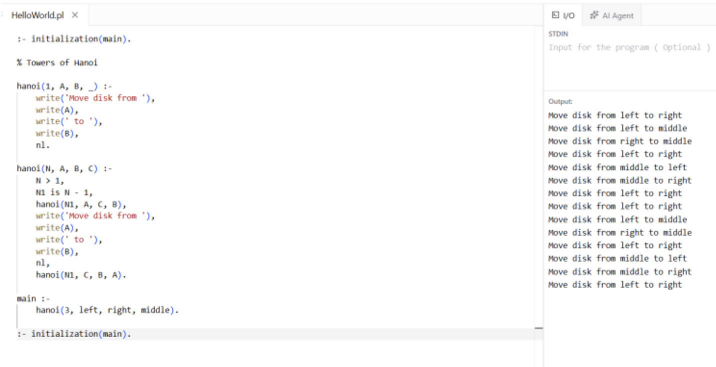
```
% Towers of Hanoi

hanoi(1, Source, Destination, _) :-
    format('Move disk from ~w to ~w~n',
        [Source, Destination]).

hanoi(N, Source, Destination, Auxiliary) :-
    N > 1,
    N1 is N - 1,
```

```
    hanoi(N1, Source, Auxiliary, Destination),
    format('Move disk from ~w to ~w~n',
           [Source, Destination]),
    hanoi(N1, Auxiliary, Destination, Source).
```

Query

?- hanoi(3, left, right, center).

Output



Result

The Towers of Hanoi problem was successfully implemented using recursion in Prolog.

Conclusion

The program demonstrates how recursive rules can be used to solve a classical problem in Prolog.